

Answers:

1. What is i^2 defined as?
 - a. -1
2. What is a complex number?
 - a. A number in the form of $a + bi$, where a and b are real numbers
3. How do imaginary numbers extend the real numbers?
 - a. They extend them by forming a 2D plane, called the complex plane, where the horizontal axis represents real numbers, and the vertical represents the imaginary numbers.
4. Why do we have imaginary numbers?
 - a. General idea is that they just make certain mathematics, like the rotations we're seeing, simpler and more elegant.
5. Write the complex number, in the form $a + bi$, for the point (1, 0) on the complex plane
 - a. $1 + 0i$
6. Write the complex number in the form $a + bi$, for the point (1, 2) on the complex plane
 - a. $1 + 2i$
7. Write the complex number in the form $a + bi$, for the point (0, -3) on the complex plane
 - a. $0 - 3i$
8. Multiply i and $1 + 2i$
 - a. $-2 + i$
9. Multiply i and $3 + 0i$
 - a. $0 + 3i$
10. Multiply i^2 and $1 + 2i$
 - a. $-1 - 2i$
11. Multiply $3 + 2i$ and $1 - i$
 - a.
$$\begin{aligned} &= (3 + 2i)(1 - i) \\ &= 3 - 3i + 2i - 2i^2 \\ &= 3 - i + 2 \\ &= 5 - i \end{aligned}$$
12. Multiply $2 + 3i$ and $2 - 3i$
 - a.
$$\begin{aligned} &= (2 + 3i)(2 - 3i) \\ &= 4 - 6i + 6i - 9i^2 \\ &= 4 + 9 \\ &= 13 \end{aligned}$$
13. Write a complex number that represents a 30° counter-clockwise rotation.
 - a. $0.866 + 0.5i$
14. What is the effect of multiplying a number by i ?
 - a. This has the effect of **rotating** that number by 90° counter-clockwise in the complex plane, the 2D plane composed of "real" numbers along X and "imaginary" numbers along Y.

Easy example: We'll use the number 1 on the complex plane, so that would be $1 + 0i$, which corresponds to a point (1, 0)

So that's $i(1 + 0i) = i + 0i^2 = i$, which corresponds to a point $(0, 1)$